SAVITRIBAI PHULE PUNE UNIVERSITY

AN AUDIT COURSE REPORT ON

"BOTNET OF THINGS"

SUBMITTED TOWARDS THE PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING

(Artificial Intelligence and Data Science)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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CERTIFICATE

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BOTNET OF THINGS

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BOTNET OF THINGS

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1. INTRODUCTION TO BOTNET OF THINGS

The Internet of Things (IoT) has changed the way people live and work. Billions of everyday objects such as smart cameras, routers, televisions, thermostats, wearable devices, medical equipment and industrial sensors are now connected to the Internet. These devices collect data, communicate with each other and perform tasks with very little human involvement. However, this rapid growth has also created a very large and often poorly protected attack surface, which cyber criminals have learned to exploit.

A botnet is a network of compromised computing devices, called bots or zombies, that are secretly controlled by an attacker known as the botmaster. When such a network is built mainly from compromised IoT devices, it is called a Botnet of Things (BoT). The owners of the infected devices usually do not notice anything wrong, because the device continues to perform its normal job while quietly obeying the commands of the attacker in the background.

Traditional botnets were built from personal computers and servers. A Botnet of Things differs because its members are low-power, always-on and Internet-facing devices that are rarely updated and are often shipped with default usernames and passwords. These weaknesses make IoT devices an attractive and easy target. The first major demonstration of this danger was the Mirai botnet in 2016, which took control of hundreds of thousands of cameras and routers and used them to launch some of the largest Distributed Denial of Service (DDoS) attacks recorded up to that time.

A Botnet of Things is generally used for the following malicious purposes:

- **DDoS attacks –** flooding a target website or service with traffic until it becomes unavailable.
- **Spam and phishing –** sending large volumes of malicious messages from many different addresses.
- **Credential stuffing and scanning –** using the bots to find and infect further vulnerable devices.
- **Proxying and anonymity –** hiding the attacker's real location by routing traffic through infected devices.
- **Data theft and espionage –** collecting sensitive information from cameras, microphones and networks.
- **Cryptocurrency mining and fraud –** using the combined computing power of many small devices for illegal profit.

Botnets are no longer only a nuisance for technical experts. They threaten businesses, hospitals, government services, critical infrastructure and ordinary households. For students of Artificial Intelligence and Data Science the topic is especially relevant, because botnet traffic detection is one of the most active areas where machine learning and data analytics are applied to cyber security.

In conclusion, the Botnet of Things is a serious consequence of the fast and often insecure growth of IoT. Understanding how it is built, how it spreads, how it is used and how it can be stopped is essential for every engineer who designs, deploys or secures connected systems.

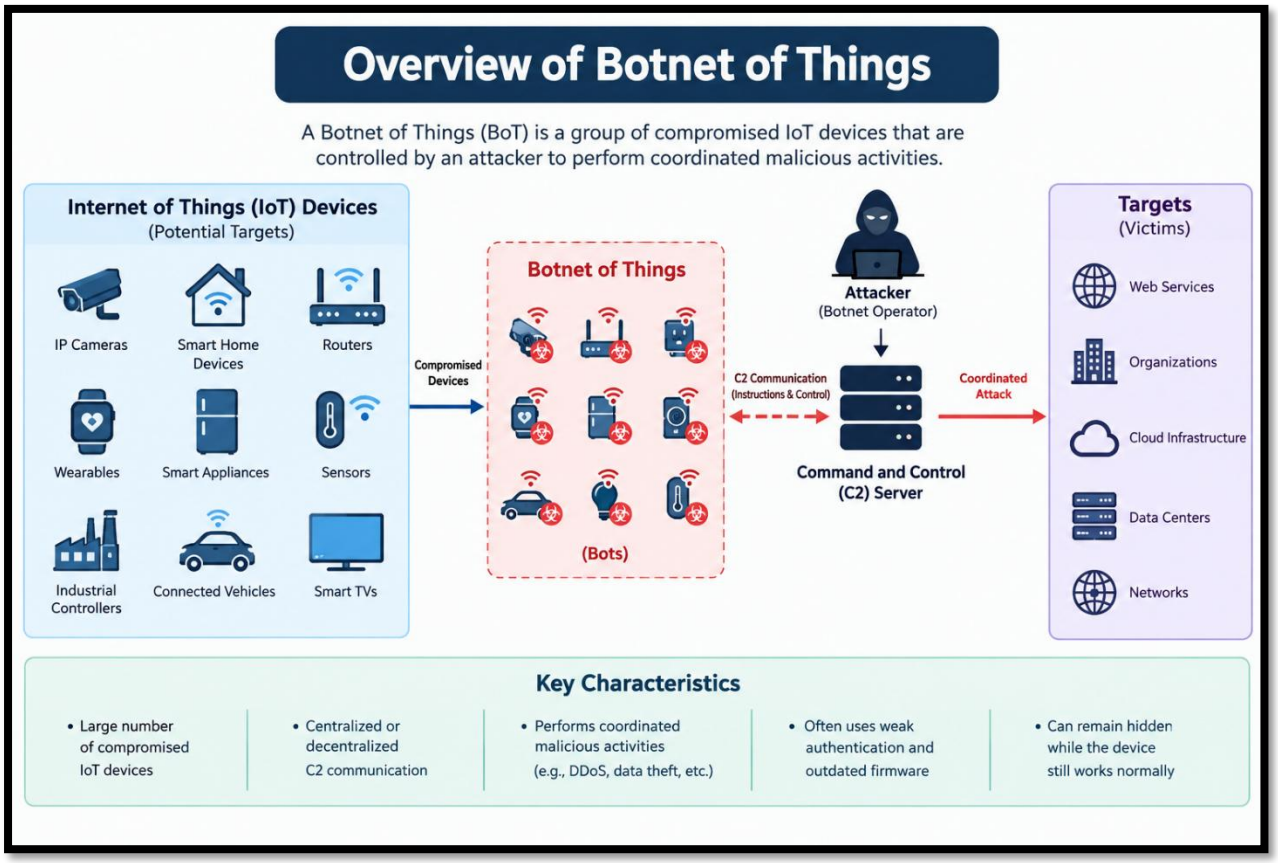


Fig.1: Overview of a Botnet of Things

2. CORE COMPONENTS AND ARCHITECTURE OF BOTNET OF THINGS

A Botnet of Things is not a single piece of software but an organised system made up of several cooperating parts. Understanding these components makes it easier to see where such a network can be detected and broken.

Botmaster (Attacker). The botmaster is the person or group that builds, controls and uses the botnet. The botmaster writes or buys the malware, decides which devices to target, issues commands and, in many cases, rents the botnet to others as a "DDoS-for-hire" service.

Bots (Infected IoT Devices). These are the compromised devices such as IP cameras, home routers, digital video recorders, smart TVs and industrial controllers. Each bot runs a small piece of malicious code that waits for instructions while keeping the original function of the device unchanged so that the owner does not become suspicious.

Command and Control (C&C) Server. The C&C server is the communication centre of the botnet. Bots report to it, receive commands from it and download updates. Because this server is the weakest point of a centralised botnet, attackers hide it using changing domain names, fast-flux techniques or encrypted channels.

Malware and Propagation Mechanism. This is the code that infects a device and then uses it to find new victims. The most common method is scanning the Internet for devices with open services such as Telnet or SSH and trying a list of default passwords. Other methods exploit known software vulnerabilities in outdated firmware.

Loader and Reporting Server. In many families, a separate loader server delivers the correct malware version for the hardware architecture of the device, such as ARM or MIPS, while a reporting server collects the list of newly found vulnerable devices.

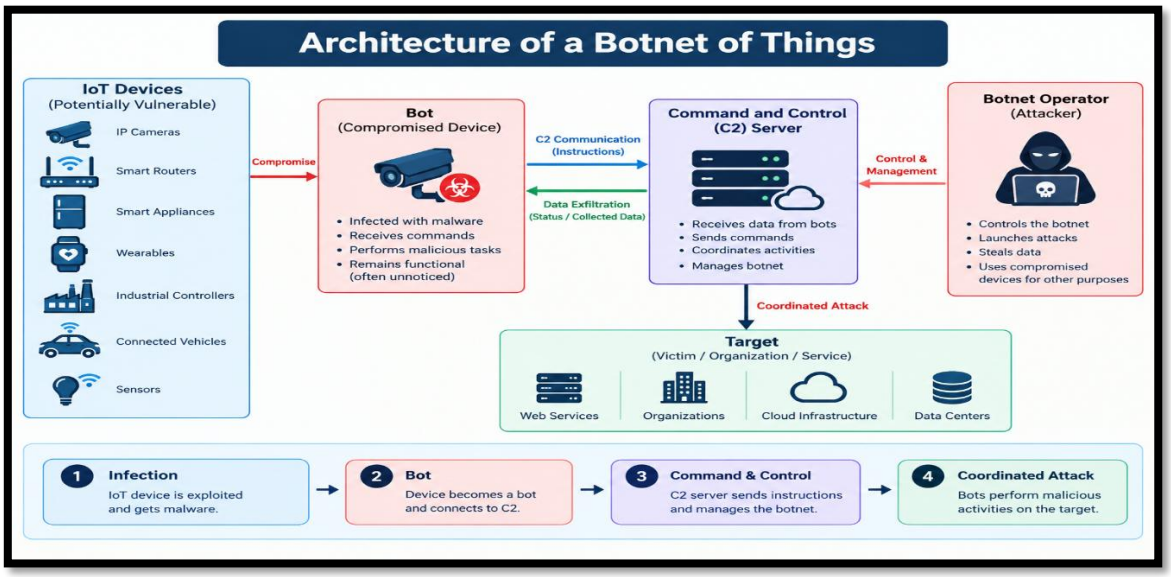


Fig.2.1: Architecture of a Botnet of Things

2.1 Life Cycle of a Botnet of Things

Most Botnet of Things infections follow a similar sequence of stages:

- **Scanning and discovery –** the malware searches for Internet-connected devices with open ports and weak security.
- **Infection –** the device is accessed using default or stolen credentials, or by exploiting a vulnerability.
- **Persistence and concealment –** the malware hides itself, removes competing malware and often blocks remote management ports.
- **Connection to C&C –** the infected device contacts the control server and waits for orders.
- **Propagation –** the bot starts scanning for new victims, which makes the botnet grow like a worm.
- **Attack execution –** on command, all bots carry out the attacker's task at the same time.
- **Update or abandonment –** the malware is upgraded with new features or the botnet is moved to a new infrastructure.

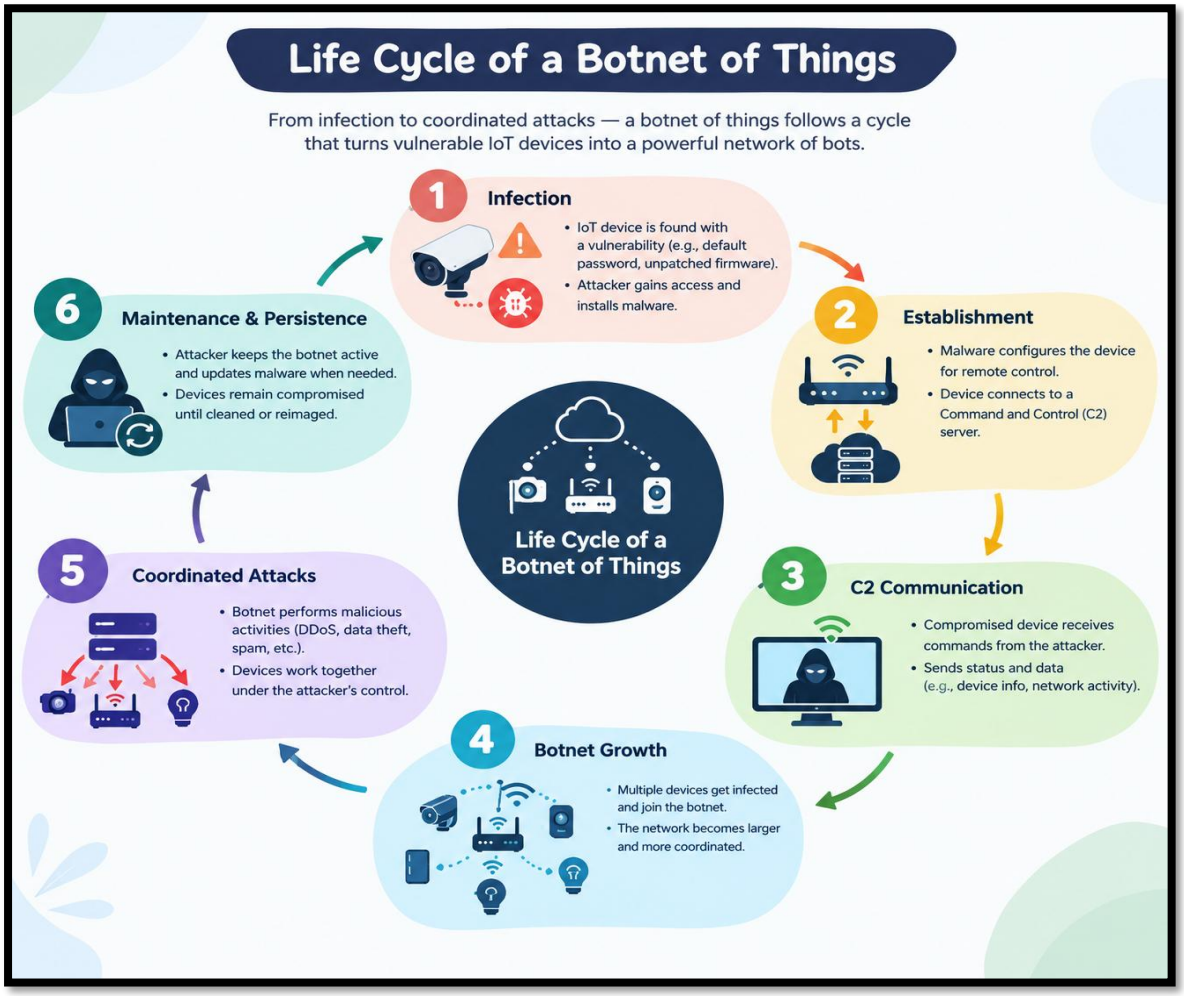


Fig.2.2: Life cycle of a Botnet of Things

3. BOTNET OF THINGS VS. TRADITIONAL BOTNETS

A Botnet of Things works on the same basic principle as a traditional botnet, but the nature of IoT devices changes its size, behaviour and impact. A traditional botnet is mostly formed from desktop computers and servers that run full operating systems and usually have antivirus software, regular patches and a human user who may notice unusual behaviour. IoT devices have none of these protections in most cases.

The main differences are summarised in the following table.

Parameter	Traditional Botnet	Botnet of Things
Type of bots	PCs, laptops and servers	Cameras, routers, DVRs, sensors, smart appliances
Operating system	Windows, Linux, macOS	Lightweight embedded Linux or firmware
Security protection	Antivirus, firewall, regular updates	Rarely any security software; weak or no updates
Credentials	User-defined, mostly stronger	Often factory default passwords
User awareness	User may notice slow performance	Devices run unattended; infection goes unnoticed
Availability	Devices are switched off at times	Always on and always connected
Scale	Thousands to hundreds of thousands	Can reach hundreds of thousands to millions
Main use	Spam, banking fraud, DDoS	Large-volume DDoS, proxying, espionage
Cleanup	Reinstall or scan with antivirus	Hard; device may need reset, patch or replacement

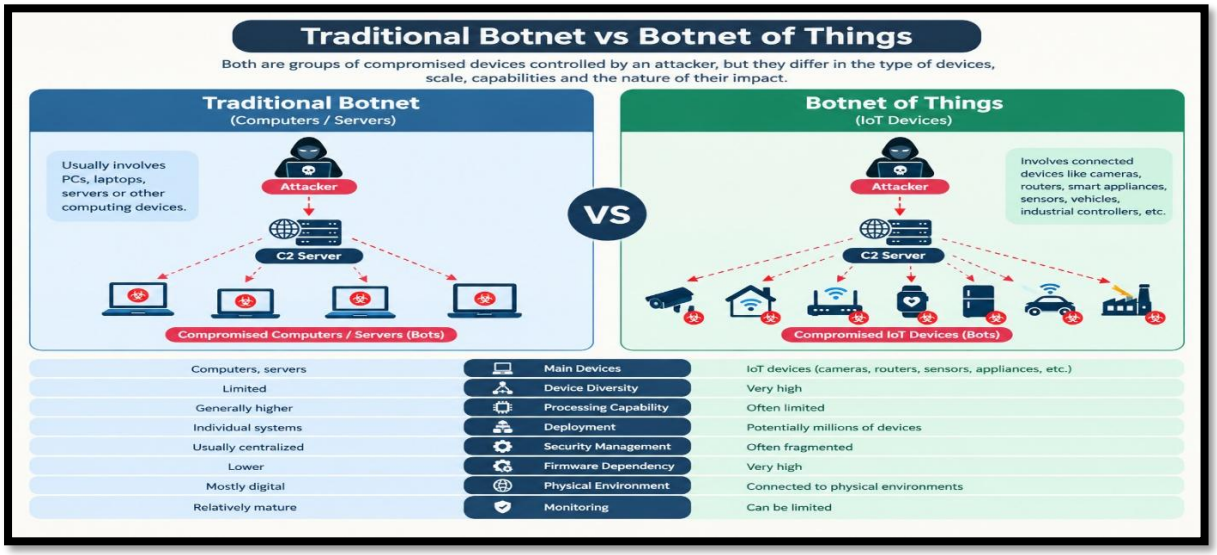


Fig.3.1: Traditional botnet compared with Botnet of Things

4. SIGNIFICANCE OF BOTNET OF THINGS

Studying the Botnet of Things is important because the number of connected devices continues to grow every year, and a large share of them are manufactured with low cost as the first priority and security as an afterthought. The danger of this situation can be understood from the following points.

Massive scale of attacks. Even a small and weak device can send a large amount of traffic when thousands or millions of them act together. Botnet-driven attacks have disrupted major websites, DNS providers and online services used by millions of people.

Low barrier for attackers. The source code of well-known botnets has been published openly, and DDoS-for-hire services allow even unskilled users to rent an attack. This makes the threat widely available.

Threat to critical infrastructure. Compromised devices are present in hospitals, factories, power systems and smart cities. An attack on these systems can affect safety as well as business.

Privacy and personal safety. Infected cameras, baby monitors and smart speakers can expose private life. Home networks can be used as a stepping stone to attack other devices.

Economic damage. Downtime, loss of reputation, mitigation costs and legal consequences place a heavy financial burden on organisations and service providers.

Hidden participation. Device owners often become unknowing participants in crime. Their devices consume bandwidth and may be linked to illegal activity without their awareness.

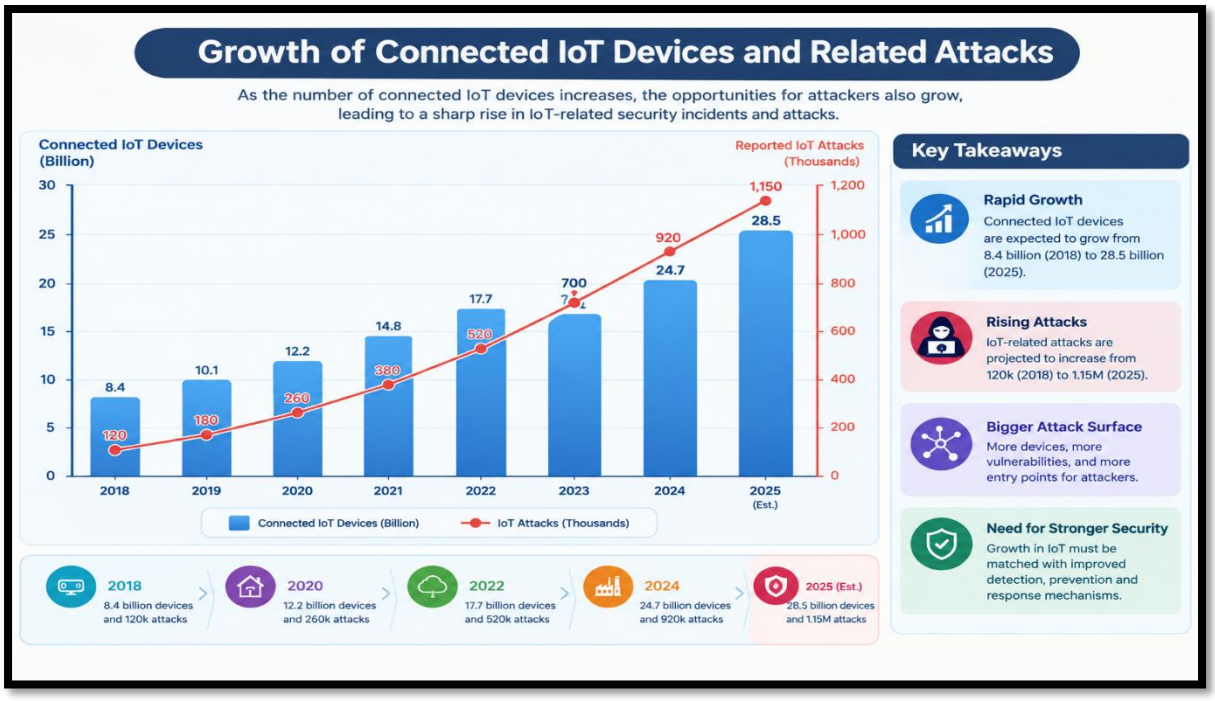


Fig.4: Growth of connected IoT devices and related attacks

5. BOTNET OF THINGS IN REAL-WORLD ENVIRONMENTS

The impact of a Botnet of Things differs from one environment to another, because each field uses connected devices in a different way. The following areas are among the most exposed.

Smart homes. Home routers, cameras, smart televisions and voice assistants are among the most common targets. Owners rarely change default passwords or install updates, so these devices are easily taken over.

Healthcare. Connected infusion pumps, patient monitors and imaging equipment are used in hospitals. Many run old software that cannot easily be patched. A compromised device can interrupt treatment or expose patient data.

Industrial and manufacturing systems (IIoT). Industrial controllers and sensors are connected to company networks for monitoring. A compromised device can be used to attack the production network or to stop operations.

Smart cities and transportation. Traffic cameras, smart meters, street lighting and parking sensors are deployed in large numbers and across wide areas, which makes monitoring and updating difficult.

Enterprise and small offices. Printers, IP phones, network cameras and small office routers are often forgotten during security audits, and attackers use them as hidden entry points.

Telecom and service providers. Large botnet attacks put heavy pressure on Internet service providers and DNS or hosting companies, who must filter malicious traffic without blocking real users.

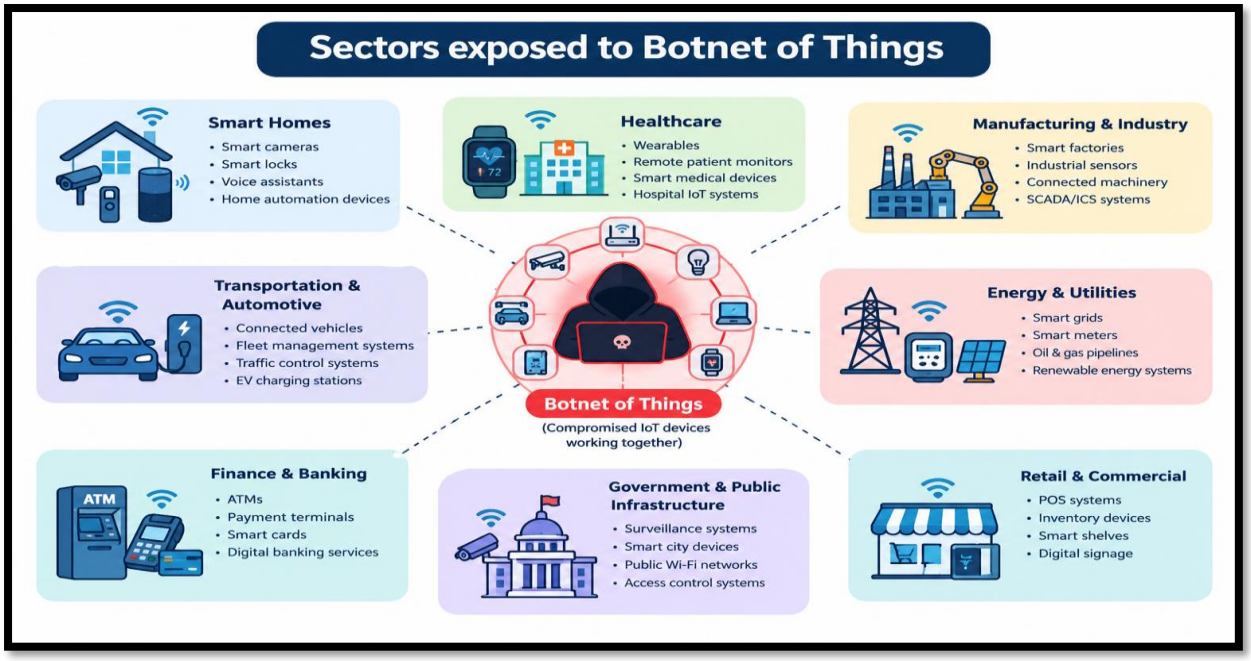


Fig 5.1: Sectors exposed to Botnet of Things

6. CHALLENGES AND CRITICISMS

Despite years of research and several large takedowns, the Botnet of Things remains a difficult problem. Many of the challenges arise from the nature of IoT itself, while others arise from the lack of responsibility and regulation.

Weak default security. A large number of devices are sold with hard-coded or default passwords, open management services and no forced password change on first use. Attackers only need to try a short list of common credentials to take control.

Limited device resources. IoT devices have small processors, little memory and low battery capacity. Heavy security tools such as full antivirus or strong encryption are often impractical, so protection must be lightweight.

Difficulty of patching. Many devices have no automatic update mechanism, and the manufacturer may stop providing updates after a short time or may no longer exist. Users may not even know that their device is vulnerable.

Heterogeneity of devices. IoT includes thousands of manufacturers, hardware types, protocols and firmware versions. A single security solution is hard to apply to all of them.

Detection difficulty. Botnet traffic can look similar to normal traffic, particularly when bots send small amounts of data or use encrypted channels. Attackers also use evasion techniques such as domain generation algorithms and peer-to-peer communication.

Lack of user awareness. Ordinary users usually do not know how to check whether a device is infected, and the device shows no clear signs. Many believe that a small device is not worth attacking.

Weak regulation and unclear responsibility. For a long time, manufacturers had no legal duty to provide basic security. Responsibility was divided among the manufacturer, the seller, the service provider and the user, so that none of them felt fully accountable.

Cross-border law enforcement. Botnet infrastructure and victims are often located in different countries, which slows down investigation and takedown.

In conclusion, the Botnet of Things cannot be solved by one technical measure alone. Overcoming these challenges requires secure design by manufacturers, regular maintenance by users, monitoring by network operators and clear rules from governments.

7. DETECTION, PREVENTION AND MITIGATION OF BOTNET OF THINGS

Protection against a Botnet of Things should follow a layered approach, in which prevention, detection and response work together. The main strategies are described below.

7.1 Prevention

- **Change default credentials –** every device should have a unique and strong password from the first day of use.
- **Keep firmware updated –** install security patches regularly and enable automatic updates when available.
- **Disable unused services –** turn off Telnet, remote management and unneeded ports, and avoid exposing devices directly to the Internet.
- **Network segmentation –** keep IoT devices on a separate network or VLAN so that an infected device cannot reach critical systems.
- **Secure-by-design products –** manufacturers should avoid hard-coded passwords, support secure updates and publish a vulnerability disclosure process.

7.2 Detection

Detection concentrates on finding unusual behaviour in network traffic or on the device. Signs of infection include sudden increase in outgoing traffic, repeated connection attempts to many random addresses, communication with known malicious domains, and unusual use of the processor or memory. Common techniques include:

- **Signature-based detection –** matching traffic or files against known botnet patterns; fast but unable to find new variants.
- **Anomaly-based detection –** learning normal behaviour and raising an alert for deviations.
- **Machine learning and deep learning –** classifying network flows as normal or malicious using algorithms such as Random Forest, SVM, LSTM and autoencoders. Datasets such as N-BaIoT, Bot-IoT and CIC-IDS are widely used for training and testing.
- **Honeypots –** deliberately exposed fake devices used to attract attackers and study their methods.
- **DNS and traffic analysis –** monitoring DNS queries and flow data for communication with command servers.

7.3 Mitigation and Response

When an infection is suspected, the device should be isolated from the network, restarted and reset to factory settings, then updated and secured with a new password before reconnecting. At the network level, service providers apply traffic filtering, rate limiting, blackholing and scrubbing services to absorb DDoS floods. Coordinated action between companies, security researchers and law enforcement agencies, such as seizing control servers and domains, has also been effective in taking down botnets.

7.4 Standards and Regulations

Several governments and organisations now publish security expectations for connected devices. Examples include NIST guidance on IoT device cybersecurity, the OWASP IoT project, the European Union's Cyber Resilience Act and the United Kingdom's Product Security and Telecommunications Infrastructure Act, which bans universal default passwords. These efforts aim to push responsibility for basic security back to the manufacturer.

8. PRACTICAL APPLICATIONS AND CASE STUDIES OF BOTNET OF THINGS

8.1 Practical Applications of the Study of Botnet of Things

Although a botnet itself is malicious, the study of botnets has many practical and positive uses in cyber security and data science.

1. Intrusion Detection Systems. Botnet behaviour is used to train machine learning models that can recognise infected devices in real time.

2. Network Monitoring and Traffic Analysis. Understanding botnet communication patterns helps administrators identify abnormal flows and unknown devices on their networks.

3. Secure IoT Product Design. Lessons from past attacks guide engineers to remove default credentials, enable secure boot and provide update mechanisms.

4. Threat Intelligence and Honeypots. Researchers deploy decoy devices to collect malware samples, attacker addresses and new techniques, which are shared with the security community.

5. DDoS Protection Services. Cloud and content delivery providers use knowledge of botnet traffic to filter attack traffic and keep services available.

6. Education and Awareness. Botnet case studies are used to train students, professionals and ordinary users on safe use of connected devices.

8.2 Case Studies of Botnet of Things

Case Study 1: Mirai Botnet (2016)

Mirai is the most widely known Botnet of Things. It scanned the Internet for devices such as IP cameras and home routers that were using factory default usernames and passwords, infected them and added them to a large botnet. In September 2016 it was used against the security journalist website Krebs on Security and the French hosting company OVH, and in October 2016 against the DNS provider Dyn. The Dyn attack made many popular services, including Twitter, Netflix and Reddit, difficult or impossible to reach for many users for several hours. The Mirai source code was later released publicly, which led to many copies and variants.

Key Learnings:

- Default passwords are a critical weakness.
- A single attack on a service provider can affect many other websites.
- Publishing malware code leads to rapid spread of variants.

Case Study 2: Mozi Botnet (2019 onwards)

Mozi was a peer-to-peer botnet that spread mainly by exploiting weak Telnet passwords and known vulnerabilities in routers and digital video recorders. Because it used a distributed hash table for communication instead of a central server, it was much harder to shut down.

Researchers reported that the botnet's activity dropped sharply in 2023 after its operators' update mechanism was disabled, but infected devices still showed that such networks can remain in the wild for a long time.

Key Learnings:

- Peer-to-peer design makes a botnet resilient.
- Unpatched routers remain a major risk.
- Cooperation between researchers and authorities is necessary.

Case Study 3: Meris Botnet (2021)

Meris was a botnet built mainly from compromised network equipment, particularly MikroTik routers. It was used to launch HTTP-based DDoS attacks at a very high request rate against several organisations, including the Russian internet company Yandex. The case showed that even professional networking equipment, if not updated, can be misused at large scale.

Key Learnings:

- Networking equipment must be patched like any other system.
- Application-layer attacks can be as harmful as bandwidth floods.
- Quick vendor response and public warnings reduce the damage.

Case Study 4: Law Enforcement Takedowns of Router Botnets (2024)

In 2024, United States authorities announced operations against botnets made from compromised small-office and home routers and other IoT devices, which were said to be used by state-linked groups to hide their activity. These actions demonstrate that Botnet of Things is no longer only a criminal problem but also a national security concern, and that technical takedown, public-private cooperation and legal action must work together.

Key Learnings:

- Old and unsupported devices are valuable to attackers.
- Botnets can be used for espionage as well as DDoS.
- Joint action by agencies and companies is effective.

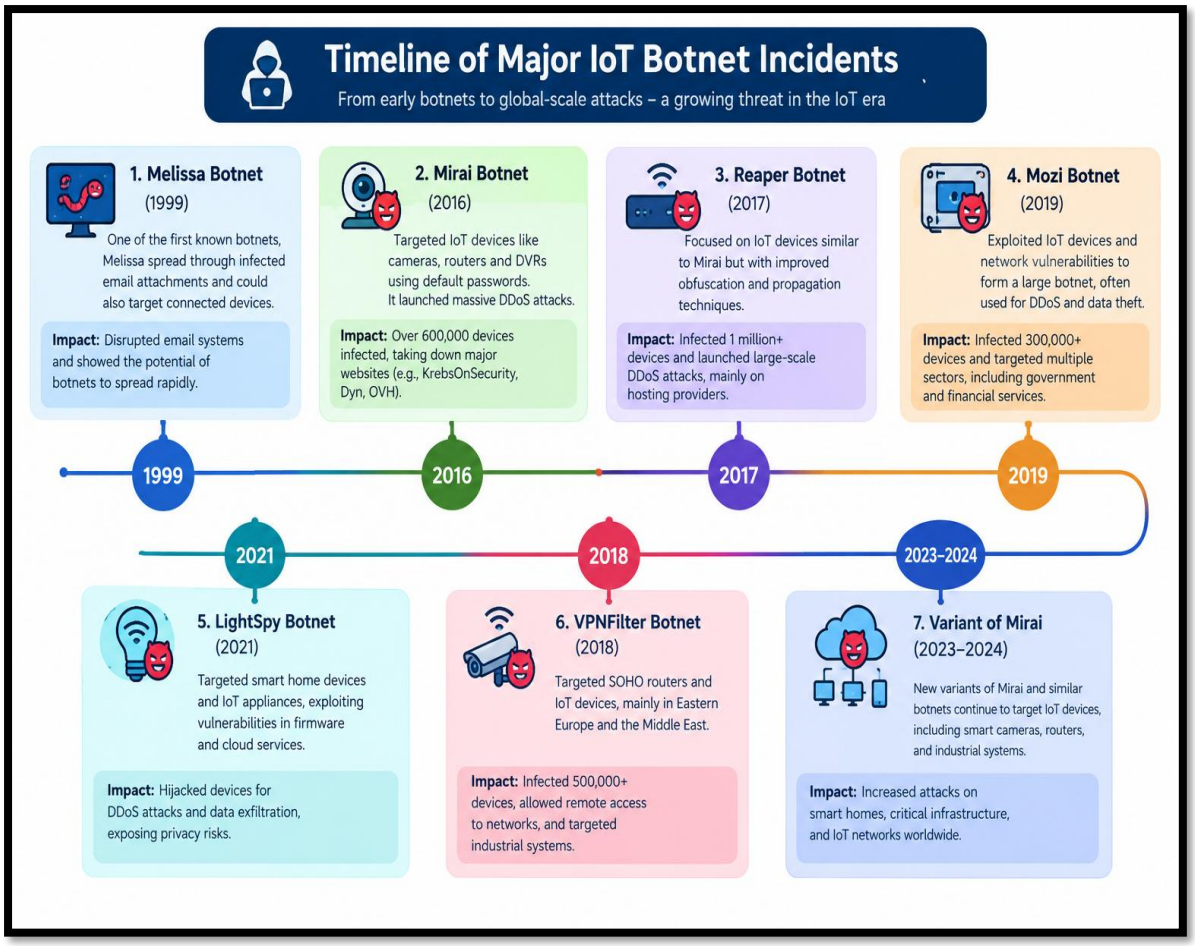


Fig.8: Timeline of major IoT botnet incidents

8.3 Conclusion

The study of the Botnet of Things has practical value in the design of safer devices, better detection systems and stronger response plans. Case studies show that the same weaknesses, such as default passwords and unpatched firmware, have been used again and again, and that they can be reduced through awareness, good engineering and cooperation.

9. COMPARISON OF BOTNET OF THINGS WITH OTHER CYBER THREATS

The Botnet of Things is closely related to other forms of cyber attack, but it has a different purpose and method. The comparison below helps to place it correctly.

Botnet of Things vs Computer Virus / Worm

A virus or worm is a program that replicates itself, whereas a botnet is a network of infected machines under central or distributed control. Many Botnet of Things families use worm-like scanning to spread, but their main aim is to keep the devices ready for remote commands.

Botnet of Things vs Ransomware

Ransomware encrypts data of the victim and demands payment, so the owner immediately knows about the attack. A botnet infection is deliberately silent and keeps the device working normally so that it can be used for as long as possible.

Botnet of Things vs DDoS Attack

A DDoS attack is a type of attack, while a botnet is the tool used to carry it out. A Botnet of Things is one of the most effective ways to create a DDoS attack, but the same botnet can also be used for spam, scanning and proxying.

Botnet of Things vs Advanced Persistent Threat (APT)

An APT is a targeted, long-term and carefully planned attack on a selected organisation. A botnet usually infects devices at random and on a large scale. However, some APT groups now use compromised IoT devices as hidden relay points, so the two threats overlap.

Parameter	Botnet of Things	Ransomware	APT
Target	Random, vulnerable IoT devices	Individuals and organisations with valuable data	Selected organisations or countries
Visibility	Silent and hidden	Visible, with ransom note	Silent and long-term
Main goal	Control of many devices for attacks	Financial extortion	Espionage or disruption
Scale	Very large	Medium to large	Small and focused
Skill required	Low to medium (code is widely available)	Medium	High

The Botnet of Things acts as a platform on which many other attacks can be built, which gives it a broader impact than most single-purpose threats.

10. FUTURE SCOPE OF BOTNET OF THINGS

As the number of connected devices continues to grow with the spread of 5G, smart cities, connected vehicles and edge computing, the Botnet of Things is expected to remain an important security concern. At the same time, new technologies are being developed to defend against it.

AI-powered attacks. Attackers may use artificial intelligence to find vulnerabilities faster, change malware to avoid detection and choose targets more effectively.

AI-powered defence. Machine learning and deep learning will play a growing role in real-time traffic analysis, anomaly detection and automatic response. Lightweight models that can run directly on edge devices are an active research area.

Growth of 5G and edge devices. Faster networks and more connected devices will allow larger botnets with higher attack capacity, so the need for network-level protection will increase.

Zero-trust security. Every device will be treated as untrusted by default and will have to prove its identity and behaviour before it is allowed to communicate.

Blockchain and secure identity. Blockchain-based device identity and secure update systems are being studied as ways to prevent unauthorised devices and tampered firmware.

Stronger regulation and labelling. More countries are expected to introduce mandatory security requirements and security labels for consumer devices, which will reduce the number of easily hacked products.

Federated learning and privacy-preserving detection. Detection models can be trained across many networks without sharing raw data, which helps in privacy-sensitive environments.

Skills and careers. There is increasing demand for professionals who understand both IoT systems and data-driven security, which is a good opportunity for AI and Data Science students.*/*

11. CONCLUSION

In conclusion, the Botnet of Things is a major cyber security challenge that has grown along with the Internet of Things. It uses the weaknesses of connected devices, such as default passwords, unpatched software and limited resources, to create large networks that can be used for DDoS attacks, spam, spying and other crimes.

The case studies of Mirai, Mozi, Meris and later router botnets show that the same mistakes are repeated and that the damage can reach far beyond the owners of the infected devices. Effective protection requires secure design by manufacturers, careful use by consumers, intelligent monitoring by network operators and strong cooperation between industry and governments.

Although the threat will continue to evolve with new technologies, the combination of good engineering practices, machine-learning-based detection, regulation and user awareness can greatly reduce its impact. For future engineers, understanding the Botnet of Things is not only a technical need but also a responsibility toward building a safe and trustworthy connected world.

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